

NORMARL: A Multi-agent Reinforcement Learning Framework for Adaptive Social Norms in Resource Sustainability

Ali Shiravand[1(✉)], Ilaria Dal Barco[2], Martino Kuntze[3], and Stefano Palminteri[1,4]

[1] École Normale Supérieure - PSL University, Paris 75005, France
ali.shiravand@ens.fr
[2] University of California Berkeley, Berkeley, CA 94720-3310, USA
[3] Sapienza University of Rome, Rome 00185, Italy
[4] Laboratoire de Neurosciences Cognitives Computationnelles (U960), Institut National de la Santé et de la Recherche Médicale (INSERM), Paris 75005, France

Abstract. Human behavior plays a crucial role in determining the sustainability of common-pool resources, where the misalignment of individual actions and collective goals often leads to overexploitation and resource depletion. Traditional methods, such as taxation and cost-based interventions, typically overlook cognitive mechanisms underlying individual decisions, limiting their effectiveness. We introduce the Norm-ORiented Multi-Agent Reinforcement Learning (NORMARL) framework, integrating psychologically plausible parameters—norm internalization and adaptive learning—to explore resource exploitation dynamics. Our model simulates agents interacting with a shared resource, adapting their behaviors based on feedback regarding normative consumption. Agents optimize their utility by balancing individual and social costs, influencing collective resource dynamics. We demonstrate that increased norm internalization and adaptive learning substantially enhance cooperation, resilience to environmental shocks, and resource sustainability. NORMARL highlights how incorporating cognitive mechanisms into computational models can reveal the potential of educational and behavioral interventions, complementing economic policies to promote sustainable behavior.

Keywords: multi-agent reinforcement learning · common-pool resources · norm internalization

1 Introduction

The sustainable management of common-pool resources represents a fundamental challenge in environmental governance, characterized by the inherent conflict between individual resource exploitation and collective sustainability. This tension, initially formalized as the *tragedy of the commons* by [5], manifests when

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rational self-interest leads to resource depletion, contradicting long-term collective benefits. While conventional approaches have advocated for either privatization or state control, recent research shows the efficacy of decentralized governance mechanisms rooted in social norms and community-based management [9,12,13,16].

Social norms—defined as shared beliefs within a group or community that influence individual behaviors [11]—serve as crucial mechanisms for promoting cooperation in resource management systems [2,3]. Recent studies have established that the effectiveness of these norms depends significantly on their internalization: the process by which norms become an intrinsic part of an individual's value system [7,17]. However, significant gaps persist in our understanding of belief updating, particularly regarding the relationship between environmental variability and the resilience of norm-based systems to external perturbations. These knowledge gaps are particularly relevant given the increasing environmental uncertainties [14–16].

Reinforcement learning (RL) presents a methodologically robust framework for examining adaptive decision-making in socio-ecological systems. Unlike traditional learning paradigms, RL enables the modeling of dynamic behavior adaptation through environmental feedback [1,18]. The integration of RL within Agent-Based Models (ABMs) facilitates the investigation of how cognitive parameters—specifically learning rates and norm internalization processes—influence both individual resource utilization patterns and system-level resilience [3,8,9].

This study advances the theoretical understanding of common-pool resources management by explicitly incorporating cognitive mechanisms into social-ecological ABM frameworks. Our research introduced the Norm-ORiented Multi-Agent Reinforcement Learning (NORMARL) simulation framework to quantify the least-studied relationship between individual learning parameters, heterogeneity, and collective resource management while evaluating the role of norm internalization in maintaining cooperative behavior under environmental uncertainty. Furthermore, we seek to identify critical thresholds for system resilience in the face of external perturbations. NORMARL can be utilized for behavioral policy design, particularly in developing educative interventions—i.e. boosting frameworks [6]—that enhance cooperation and system resilience. By elucidating the mechanisms through which cognitive processes influence collective behavior, this research contributes to both theoretical understanding and practical management of social-ecological systems under uncertainty.

2 Methods

2.1 Baseline Model Without Social Learning

First, we simulated a baseline model with a pool of agents without learning. Agents deriving individual utility from resource consumption as $U(Q) = \sqrt{Q}$, incurring an individual extraction cost $C(Q) = Q$, where Q is the quantity consumed. By maximizing $U(Q) - C(Q)$, agents identify an optimal selfish action $Q^* = 0.25$ (highlighted in Fig. 1b). Resource consumption by agents affects the

shared resource stock E, that evolves as:

$$E_{t+1} = (1+\delta).E_t - \sum_{i=1}^{N} Q_t^i, \tag{1}$$

where $\delta = 0.02$ represents the natural regeneration rate, N is the number of agents, and Q_t^i is the consumption by agent i at time t. The term $(1+\delta)E_t$ accounts for resource replenishment, while $\sum_{i=1}^{N} Q_t^i$ represents the collective consumption. As simulated results show in Fig. 1e, when agents act selfishly ($Q^i = 0.25$ for all agents) without considering social costs, resources are rapidly depleted. This demonstrates the necessity of social mechanisms to regulate behavior and ensure sustainability.

2.2 NORMARL—A Belief Updating Model for Learning Norms

Then we developed the NORMARL framework to study the role of learning social norms in sustainable resource consumption. To discourage unsustainable consumption, a social cost function $S_i(Q_t^i)$ is introduced:

$$S_i(Q_t^i) = \begin{cases} 0 & \text{if } Q_t^i < \bar{Q}_t^i, \\ \beta_i.(Q_t^i - \bar{Q}_t^i).\max\{0, (a - b.E_t)\} & \text{if } Q_t^i \geq \bar{Q}_t^i, \end{cases} \tag{2}$$

where $\bar{Q}_t^i$ is the perceived social norm (belief about average consumption in the society), β_i measures the strength of norm internalization for the ith agent, a and b are positive constants, and E_t is the current resource stock. Social costs penalize consumption above the agent's perceived norm, proportional to the internalization factor and deviation from the perceived norm, and inversely proportional to the resources left.

Figure 1a illustrates the interaction loop, where agents consume a shared resource and update their beliefs about societal norms based on environmental feedback. Here in the new belief updating model, the agent integrates individual and social values into decision-making. The value of an action (V_i) combines individual utility, individual extraction cost, and social cost (Fig. 1d):

$$V_i(Q^i) = U_i(Q^i) - C_i(Q^i) - S_i(Q^i), \tag{3}$$

and agents select actions probabilistically using a softmax function from a discrete space $\{0.05, 0.10, \ldots, 0.40\}$ (Fig. 1c):

$$P(Q^i) = \frac{e^{V_i/\gamma}}{\sum_{k=1}^{|A|} e^{V_k/\gamma}}, \tag{4}$$

where γ is fixed and controls the exploration-exploitation trade-off and $|A| = 8$ (see Fig. 1c). Agents update their perception of the social norm $\bar{Q}$ using a weighted average of their prior belief and the observed average consumption ($\bar{q}_{t+1}$ - feedback by the environment to each agent on each trial):

$$\bar{Q^i}_{t+1} = (1-\alpha_i).\bar{Q}_t^i + \alpha_i.\bar{q}_{t+1}, \tag{5}$$

where α_i is the learning rate for agent i. A higher α allows faster adaptation to changes in societal consumption, while a lower α reflects slower learning. Simulations were conducted with $N = 100$ agents and an initial resource stock $E_0 = 1000$. The analysis explores how variations in β, α, and heterogeneity among agents affect resource sustainability.

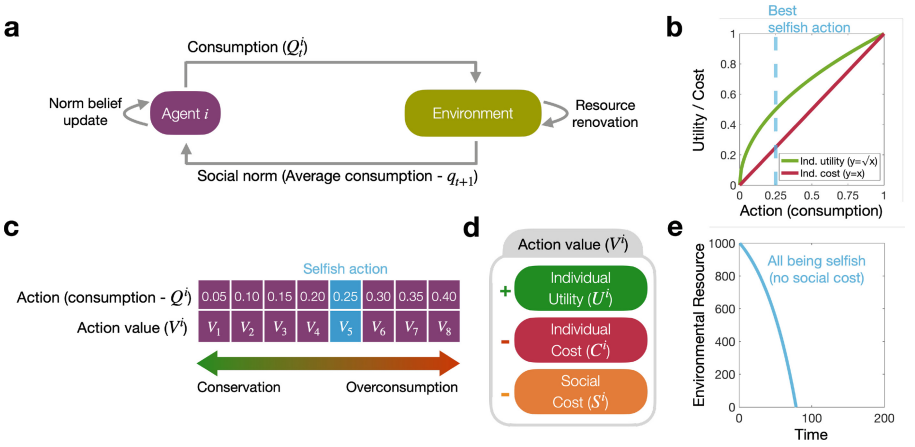


Fig. 1. MARL simulation framework. **a**, Interaction diagram between an agent and the environment. **b**, Utility/cost function and the best selfish action of an agent without social cost. **c**, Action space of an agent. **d**, Building components of an action value. **e**, Quick depletion of resources when all agents act selfish without social cost.

3 Restuls

We investigated the role of norm internalization (β) and learning rate (α) in resource sustainability, welfare, and system resilience to environmental shocks. Welfare, defined as the average consumption across all agents in a given trial, is depicted in Figs. 2 and 3 using colormaps (warmer colors indicate higher welfare levels).

3.1 Impact of Norm Internalization and Learning Rate

Norm internalization, represented by β, significantly influences resource sustainability and welfare. As shown in Fig. 2a, higher internalization levels ($\beta = 1.0$ for all agents) lead to sustainable resource dynamics, while lower levels ($\beta = 0.6$ for all agents) result in rapid resource depletion. The results suggest the existence of a critical threshold for β; below this threshold, resources are consistently overexploited, and sustainability cannot be achieved.

The colormap in Fig. 2a reveals that higher β values not only stabilize resources but also maintain higher welfare levels (other parameters are fixed). This alignment between individual and collective benefits highlights the role of internalization in ensuring both ecological and societal well-being. However, the presence of non-internalizers (NI - those who have $\beta = 0$) within the population introduces significant challenges. When the NI ratio exceeds 20%, even moderate β values fail to sustain the resource, leading to inevitable depletion.

The learning rate (α) controls how quickly agents adapt their beliefs concerning the average level of consumption based on the feedback from the environment. In Fig. 2b (left panel), higher learning rates ($\alpha = 0.09$ for all agents) enable agents to maintain resource levels and the converged equilibrium is independent of the values of α (other parameters are fixed). In contrast, lower learning rates ($\alpha = 0.005$ for all agents) lead to delayed adaptation and therefore resource depletion.

The right panel of Fig. 2b highlights how the proportion of slow-learners (SL - those who have very small α) influences sustainability. Although systems remain stable with lower SL ratios, increasing SL ratios disrupt adaptation and accelerate resource collapse, regardless of higher learning rates among the remaining population.

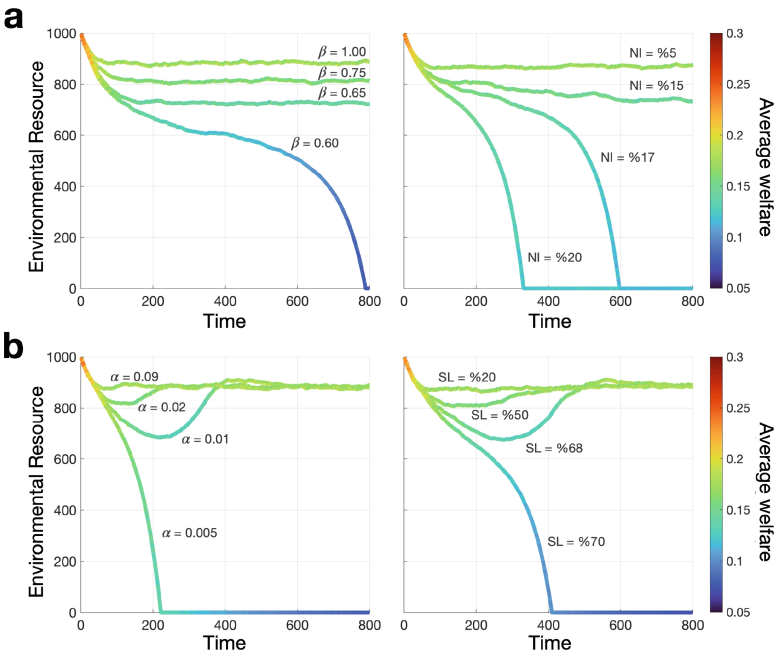


Fig. 2. Simulation results. **a**, Resource dynamics and welfare for different internalization factors (β) and ratios of non-internalizers (NI). **b**, Resource dynamics and welfare for different learning rates (α) and ratios of slow-learners (SL).

3.2 System Resilience to Environmental Shocks

To assess system resilience, an environmental shock was induced at trial 450 (which is after the stabilization of the system), reducing the resource stock abruptly. Figure 3 highlights the recovery dynamics under varying α and β values, providing critical insights into the adaptive capacity of the system.

In Fig. 3a, higher learning rates ($\alpha = 0.30$) enable agents to adapt quickly to the shock, restoring resource levels and maintaining higher welfare. Conversely, lower learning rates result in delayed adaptation, prolonged resource instability, and significant welfare losses. Similarly, Fig. 3b demonstrates that higher internalization levels ($\beta = 1.50$) enhance resilience by fostering cooperation and stabilizing resource use post-shock. In contrast, lower β values ($\beta = 0.75$) fail to prevent resource collapse, underscoring the critical role of internalization in ensuring system recovery.

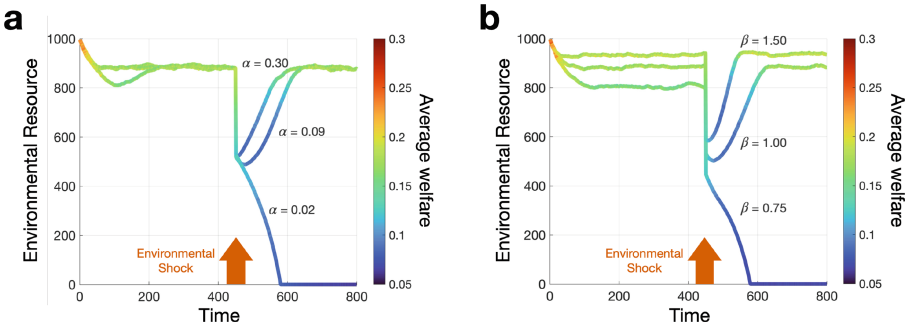


Fig. 3. Effects of an environmental shock (induced at $t = 450$) on the resource and welfare, to evaluate the resilience of the system under different parameters. **a**, Resource dynamics for varying learning rates (α) and **b**, internalization factors (β).

4 Discussion

Our NORMARL simulations underscore the crucial roles of norm internalization (β) and learning rates (α) in fostering sustainability and resilience. Elevated β values enhance cooperation by aligning individual behaviors with societal norms, while increased α enables agents to adapt swiftly to changing conditions, ensuring robust recovery from environmental shocks and sustained welfare levels.

Traditional environmental policies often emphasize taxation and cost-related interventions to curb overexploitation [10]. While effective in certain contexts, these measures may overlook the cognitive and behavioral processes driving decision-making [4]. Our findings highlight the significance of educational and behavioral interventions that enhance norm internalization and adaptive learning. Grounded in an understanding of human behavior, these strategies

address the root causes of unsustainable practices, fostering long-term cooperative behaviors.

It's essential to acknowledge the inherent limitations of computational models used in this study. The selection of hyperparameters can significantly influence model outcomes if not appropriately calibrated. Recognizing these limitations is crucial for effectively applying computational tools in policy-making [14]. By emphasizing the interplay between cognitive parameters, social dynamics, and environmental resilience, this work provides a framework for understanding and enhancing the mechanisms underpinning sustainable resource use. These insights offer actionable guidance for designing interventions that promote sustainability in complex socio-ecological systems.

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